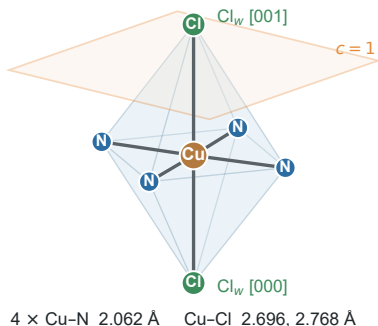


(A) XUMMID: boundary-spanning CuN_4Cl_2



(B) One wrapped Cl site, two edges

donor	τ	d / Å
Cl_w	(0,0,0)	2.696
	(0,0,+1)	2.768
τ omitted	CuN_4Cl SPY	CN = 5 $S(\text{SPY}) = 1.0100$
τ retained	CuN_4Cl_2 O_h	CN = 6 $S(O_h) = 1.8748$

(C) CSD MOF periodic-state census

172,332 local states · 36,552 metric IDs · 60,139 stereo-distinct IDs

Processed

15,905 / 15,906 **99.99%**

State-emitting entries

10,948 / 15,906 **68.83%**

Entries with $\geq 1 \tau \neq 0$ site

9,805 / 10,948 **89.56%**

Sites with $\tau \neq 0$

74,354 / 172,332 **43.15%**

(D) Same periodic state after crystal re-expression

atom/donor order · cell origin · periodic image/orbit · $\det(U) = +1$ basis

entry-level records	400 / 400
metric local-state IDs	5,920 / 5,920
stereo local-state IDs ^a	5,920 / 5,920
reflected-cell metric IDs	50 / 50

^aStereo identity was tested for orientation-preserving basis changes.

0 50 100%

0 50 100%
identity retained